Ajit Singh

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SKILLS:

Programming Language: C++, STL, C#, HTML, CSS, JavaScript, python(minor) |**DevOps Tools:** Git, TFS, WinDbG
Other Tools and knowledge: SolidWorks**, Visual Studio Profiling, Windows programming using Visual studio, json, XML**
Conceptual Knowledge: OOPs, **Design Patterns**, SOLID Principles, **Data structure and algorithms, multi-threading, Operating system**

EDUCATION

MTech: **Indian Institute of Technology (IIT), Ropar (Manufacturing Engineering)** **2021 |CGPA: 7.4**
BTech: Punjab Technical University (Mechanical Engineering) **2018 | 74.76%**

TOTAL EXPERIENCE: **4.5+ YEARS**

1-SIEMENS | Software Engineer | April 2026 – Present| Pune, India

Contributed to the development and maintenance of **Remote Data Access** (RDA) device software used in charger locomotives across the European region.

1. Contributed to the **development, debugging, and maintenance** of embedded software within an existing **C/C++ codebase.**
2. Performed **software testing and validation** to verify bug fixes and ensure expected application behavior.
3. Worked on **upgrading the OpenVPN version** in the RDA software environment.

2-DASSAULT SYSTEMES | R & D Software Engineer Specialist| March 2022 – Nov 2026| Pune, India

Worked as a part of Dassault Systemes, Power BY team, SolidWorks.

1. **Cloud Reload Optimization:** Cut cloud reload latency **by 80%** via **LRU-based caching** and distributed file sync, improving global user performance in SOLIDWORKS assemblies**. [Tech: C++]**
2. **BATCH SAVE SCHEDULER:** Designed a **Graph-Dijkstra-algorithm** based scheduling algorithm to predict batch completion time (**90% accuracy**) and enhance report refresh for large CAD files. [**Tech: C++17, C#, STL]**
3. **CLOUD FILE HANDLING SYSTEM (C# Web Services):** Built a zero-installation cloud drive (Upload, Download, Drag-and-Drop), eliminating the need for local setup and saving ~40 minutes of installation time, enhancing collaboration. [**Multithreading, C++, C#**]
4. **Automation Tool for Save/Reload Testing**: Developed an internal test executable for the team that loads locally built DLLs and invokes key functions like "*save"* and "*reload"* post-code changes, improving validation and reducing manual testing.

PERSONAL PROJECTS

| LinguaVoice | **A Chrome Extension**: Built to reduce language barriers.
Live and published on **Chrome web Store** - 10 September, 2024, Pune, India
Features:

- Developed a multilingual tool with automatic language detection **(~90% accuracy**) and real-time text translation.
- Implemented **voice input/output** and command recognition to enable spo ken translations.
- Integrated OCR to extract text from images with **~95% accuracy** and support translation.
- Secure Google Drive integration via **Google OAuth 2.0**. **[Technologies used: Built with HTML, CSS, and JavaScript]**

CERTIFICATION & Subjects Studied

- **Coding for Everyone in C/C++** - Udemy, **|Operating System |Data Structures and Algorithms** - Udemy**, |ML Algorithms**
- **System Design** – Udemy-covered concepts like Load-Balancing, Content Delivery Network (CDN) in details

CODING HACKATHONS

- **Runner-up** in Dassault System **Hackathon** 2023 - Presented" Magical Voice" **idea**: a new way to design small components using voice commands in SolidWorks.